INCH-POUND

MIL-P-60620D (AR)
AMENDMENT 1
23 August 1994

MILITARY SPECIFICATION

PROJECTILE ASSEMBLY FOR
CARTRIDGE, SMOKE, WP, 81MM, M375A1

This Amendment forms a part of MIL-P-60620D (AR), dated 9 August 1985, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1.1, SPECIFICATIONS, Add:

"FEDERAL
TT-C-490 Cleaning Methods and Pre-treatment of Ferrous Surfaces for Organic Coatings."

PAGE 4

3.6, Delete in its entirety and substitute:

"3.6 Proving ground (maximum average pressure) test. Projectile bodies shall reveal no evidence of failure when tested at maximum average pressure in accordance with 4.5.6."

PAGE 5

Add new paragraph 3.9 as follows:

"3.9 Hydrostatic test (one-piece projectile). Each one-piece projectile shall withstand stresses developed by the application of 7000 ± 250 psi internal hydrostatic pressure for five seconds without leakage, rupture, or permanent deformation when tested in accordance with 4.5.2.1."

AMSC N/A
FSC 1315
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PAGE 5 (cont)

4.3.1, List of components submitted for first article, Add:

"Body, prior to machining obturator groove ...10535864 ...10".

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TABLE 1, Add Category IV as follows:

"Body, prior to machining obturating band groove (Dwg. 10535864)

a. Examination for defects ...10 ...100% ...3.2 ...4.4.2.2/4.4.2.3/4.4.2.4

b. Hydrotest ...10 ...100% ...3.9 ...4.5.2.1".

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4.4.2, Add the following at the end of the paragraph:

"Dimensions marked tool control (*) shall be gaged during first article inspection, at the beginning of each workday, every four (4) hours during production, for the last component prior to and first component after each tooling, equipment, or process change, repair or adjustment. Measurements of the tool may be substituted provided the contractor has established correlation between the tool dimension and the component dimension prior to substitution. When acceptability has been determined, inspection of tool dimensions will be permitted. Whenever a defective tool control dimension is found, immediate action will be taken by the contractor to correct the tool problem and all material produced since the last successful inspection shall be rejected."

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4.4.2.1, Majors, Add:

"103 Forward inside diameter ...0.40% ...3.2 ...Gage
104 Wall thickness, minimum, through shoulder ...0.40% ...3.2 ...Gage
105 Length from rear face to base of cavity ...0.40% ...3.2 ...Gage"

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4.4.2.2,

Title block, Add: "Body".

Drawing Number block, Add: "10535864".

Notes, Add:

"1/ Minor characteristics do not apply to 10535864".

Critical, Add: "4 ...Hydrotest ...100% ...3.9 ...4.5.2.1".

Critical 2, delete in its entirety and substitute:

"2 ...Press fit surface increases in diameter anywhere from front face to WP cavity ...100% ...3.2 ...Gage."

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4.4.2.3,

Critical 1, Delete in its entirety and substitute:

"1. Hydrotest ...1/ ...100% ...3.4.1 ...4.5.2".

Critical 2, Delete in its entirety and substitute:

"2. Air leak test ...1/ ...100% ...3.4.2 ...4.5.3".

Criticals 3, 4 and 5, Delete in their entirety.

Specials, Add:

"a. Metal defects ...100% ...3.2 ...Visual
b. Hydrostatic test stamp missing ...100% ...3.2 ...Visual
c. Air pressure test stamp missing ...1/ ...100% ...3.2 ...Visual
d. Braze metal not visible for 360 degrees, internal and external ...1/ ...100% ...3.4 ...Visual
e. Braze metal voids in excess of requirements (external) ...1/ ...100% ...3.4 ...Visual
f. Presence of void(s) at juncture of base and body wall (internal) ...1/ ...100% ...3.4 ...Visual
g. Joint quality inadequate ...1/, 2/ ...100% ...3.4.3 ...4.5.4/Visual".

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Notes, Add:

"1/ Alternative body assembly only

2/ See 4.4.3.4".

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4.4.2.3, Majors 110, 111, and 112

No. of Sample Units column, Add: "1/".

Notes, Add:

"1/ See 10535866 for this feature on Body Alternative I."

Majors, Add:

"120 Diameter of bourrelet ...0.40% ...3.2 ...Gage".

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4.4.2.3,

Majors 113, 114, 120, and Minors 201, 202, 203, 204

No. of Sample Units column, Add: "1/".

Majors 115 and 116

No. of Sample Units column, Add: "2/".

Notes, Add:

"1/ See 10535866 for this feature on Body Alternative I

2/ Applies to 10535864 only".

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4.4.2.3, Minors

210 No. of Sample Units column, Add: "1/".

211 Delete: "Visual", and substitute: "Gage".

218 Delete in its entirety and add:

"219 Depth of drill point in thread boss, maximum ...0.65% ...3.2 ...Cage".

220 Diameter of drill point at rear face of thread boss ...0.65% ...3.2 ...Gage".

Notes, Add:

"1/ See 10535866 for this feature on Body Alternative I."

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4.4.2.3, Add:

"Special a ... Major diameter of thread, min ... 100% ...3.2 ...Gage".

Major 102, delete in its entirety.

Major 104, No. of Sample Units column, Add: "1/".

Majors, Add:

"111 Salt spray test ...2/ ...- ...3.5 ...4.5.5/Visual
112 Max average pressure test ...3/ ...- ...3.6 ...4.5.6/Visual".

Notes, Add:

"1/ Major diameter shall be checked to full form minimum length
2/ See 4.5.5
3/ See 4.4.4".

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4.4.3.2, Delete in its entirety.

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4.4.3.4, Delete in its entirety and substitute:

"4.4.3.4 Joint quality (alternative body assembly). The test shall be performed for first article inspection each day of brazing during production and whenever a change is made to the brazing process. After air and hydrostatic testing, but prior to the application of protective coating, three (3) samples from the start of brazing operations and from the first units produced after a layoff of over one (1) hour shall be tested. Two (2) samples shall be selected, each from the end of brazing operations during continuous brazing, up to 24 hours and from the last units produced prior to a layoff of over one hour and tested in accordance with 4.5.4. If a failure occurs in either of the two (2) samples tested at the end of the day's brazing operations or at the start of layoff, all parts brazed since the last successful test shall be rejected."

4.4.3.5, Delete in its entirety.

4.4.4, Delete in its entirety and substitute:

"4.4.4 Proving ground (maximum average pressure) test. See 4.5.6. Twenty-five (25) projectiles shall be tested from the first article sample and ten (10) from each of the first three (3) production lots and from three (3) consecutive lots when a new mill heat of steel is introduced into production. An attempt will be made to recover at least 60 percent of the projectiles fired in each test. If a round falls short due to an undetermined cause and is not recovered, it shall be classed defective. Defect classification and individual accept/reject criteria shall be in accordance with Table II.

Table II - Proving Ground Test

Category	Defect	Accept/Reject	
Critical	Premature burst, metal part breakup or deformation due to launch forces	0	1
Major	Horizontal range less than 80% of average range	0	1".

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Add new paragraph 4.5.2.1 as follows:

"4.5.2.1 One-piece hydrotest. After the wall thickness is in conformance with final size, but prior to machining the obturating band groove, each one-piece projectile shall be subjected to an internal hydrotest to determine compliance with the requirements of 3.9. A temporary internal thread shall be machined in the forward end of the body to couple the body to the hydrostatic test fixture. Only the threads and front face of the projectile body may be in contact with the test fixture. After the required minimum pressure is met or exceeded, the pressure shall be held for five (5) seconds, minimum. Projectile rupture, cracking leakage or bulging during the test shall be cause for rejection of the body.

If more than one projectile is tested simultaneously on a manifold and one projectile fails the test, it shall be replaced with a bew body. All other projectiles attached to the manifold at that time shall be retested with the new body until all simultaneously meet the time and pressure requirements.

Immediately after removal from the hydrotest equipment, each projectile that has passed the hjydrotest shall be stamped "H" 1/8 inch high by .01 inch deep approximately 1/8 inch below the location of the zone used for the lot number.

The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over four (4) hours of continuous operation. It shall also be calibrated prior to restarting the operation after a layoff of three hours or more. If it has been found that the equipment is out of calibration in such a direction that the pressure was actually lower than required by the test, the bodies tested since the last satisfactory calibration shall be retested after the calibration error has been corrected.

Protective barriers should be installed where applicable to shield operators and material from metal fragments which may come from defective projectiles that fail the test. Examination should be made of all hydrotest plugs after a projectile is blown during testing. Badly nicked or damaged plugs may cause destruction of acceptable items. The protective barrier must allow sufficient clearance for a defective projectile to completely fail in any direction."

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4.5.5, Delete in its entirety and substitute:

"4.5.5 Salt spray test. In performing the following tests, TT-C-490 shall apply unless specified otherwise, herein. The test shall be performed for first article inspection, each day of painting during production and whenever a change is made to the paint process, until all samples from five (5) consecutive days of painting have passed the test. Thereafter, the sample frequency may be reduced to twice per week during continuous days of painting. Five specimens shall be selected, consisting of randomly chosen projectiles which have been processed through all the same cleaning, phosphating, painting and drying steps, at the same time using the same equipment as other production projectiles. The samples shall be taken from each continuous final paint operation, up to 24 hours long. One projectile shall be taken during the first thirty (30) minutes of painting, one during the last thirty (30) minutes, and the others at any other time.

Prior to salt spray testing, two (2) of the specimens must only pass an adhesion test using the procedures and pass-fail criteria contained in 4.2.8 and 3.5.6, respectively, of TT-C-490. If both specimens pass the adhesion test, the three (3) remaining specimens shall be scored lengthwise using a precision scribing tool/insert or equivalent in preparation for the salt spray test.

The salt solution, test equipment, and drying procedures shall be in accordance with ASTM B117, Salt Spray (Fog) Testing, unless specified otherwise, herein. Specimens shall be positioned between 15 and 30 degrees from vertical during the test. No coating of any kind shall be permitted over the paint on stamp marks of the specimens prior to testing. However, areas in contact with racks or supports, unpainted surfaces and areas within 3/16 inch of unpainted surfaces may be protected.

When examining the specimens for signs of apaint failure, the entire surface of the projectile shall be considered the significant surface.

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As used herein, the term "blister" refers to an unbroken area of paint with a bubble-like appearance, whereas "pits" or "spots" are exposed phosphate pretreatment or bare metal visible through holes in the paint. Blisters which are present when the test is over are to be counted within five (5) minutes of removal from the test chamber.

Projectiles shall be tested for the duration specified on drawing 10522511. After testing, any projectile which fails for any of the following reasons shall cause rejection of all parts painted since the last successful test and a return to the original sampling plan if failure occurred during reduced sampling. The failure criteria a - i apply to the significant surface only.

a. The paint shows more than 1/8 inch creepage, blistering or loss of adhesion of paint from the scribe mark, stamp mark, or edge.

b. There are more than eight (8) blisters, total.

c. Any blister exceeds 1/16 inch average diameter.

d. Any pit exceeds 1/8 inch average diameter.

e. There are four (4) or more scattered pits exceeding 1/16 inch average diameter.

f. There are four (4) or more spots within one square inch.

g. There are three (3) spots within one square inch in two (2) or more places.

h. There are more than six (6) scattered spots, total.

i. There is evidence of rust under the paint (a brown discoloration showing through the paint) in excess of 1/4 inch in diameter total area.

j. There is some variation of the above or an unlisted defect anywhere which the government representative deems excessive."

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4.5.6, Delete in its entirety and substitute:

"4.5.6 Proving ground (maximum average pressure) test. The samples supplied in accordance with 4.4.4, inert loaded to the service weight and dummy fuzed, will be assembled into complete rounds and fired for recovery from a mortar for which the round is standard, the tube of which will have a minimum of 25 percent life remaining based on current mortar tube condemnation criteria. The propelling charge will be adjusted to produce 10,028 - 10,396 psi with cartridges conditioned at 70 degrees F. The test shall be performed at a government proving ground in accordance with applicable U.S. Army TECOM acceptance test procedures. Photographic equipment will be set up near the muzzle to record the flight of the projectile to help determine compliance with 3.6 and Table II in 4.4.4. The projectiles shall be recovered and examined to determine compliance with 3.6."

PAGE 21

6.3, Delete: "Dover, N.J. 07801-5001.", and substitute:

"...Picatinny Arsenal, NJ 07806-5000."

6.4, Delete: "Dover, N.J. 07801-5001.", and substitute:

"...Picatinny Arsenal, NJ 07806-5000."

Custodian:
Army-AR

Preparing activity:
Army-AR

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